ORCA: Orbital Recycling and Construction Array

Catalog, Sensor, and Resource Limits on Reusable Retrieval of SL-16 Rocket Bodies

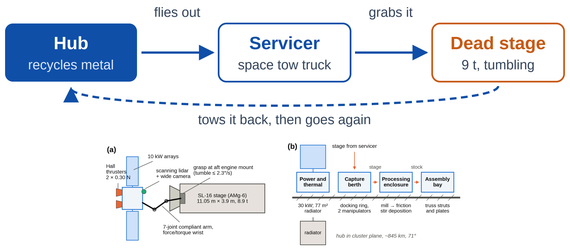
Amey Mishra · Harmony School of Advancement, Houston, Texas · Accepted to IEEE ISMSIT 2026 (publication pending in IEEE Xplore)

1 The problem

Dead rocket stages are the most dangerous junk in orbit

- A crash between two could create thousands of fragments that threaten working satellites
- **18 of the 50** most dangerous objects are the same model: Soviet **SL-16** upper stages [1]
- Each is **11 m long**, weighs **9 t**, and drifts dead at **~850 km** altitude [6]
- **Why it matters:** collisions could make key orbits unusable for weather, GPS and communication satellites

2 The idea



- A reusable **servicer** (space tow truck) tows back one stage per trip: **one spacecraft, many stages**
- A **hub** recycles the aluminum into construction beams

Reference design

Servicer dry mass / xenon	1,350 kg / 450 kg
Two Hall thrusters: thrust / I_{sp}	0.60 N / 1,900 s
End-of-life power / capture capacity	10 kW / 9 t

$$P_{\text{jet}} = \frac{1}{2} F g_0 I_{\text{sp}} \approx 5.6\ \text{kW}$$

3 The question

Would this actually work, and if not, what would stop it?

- Past studies check one piece at a time (route, navigation or recycling) and assume the rest works [2]
- This study tests **the whole mission chained together**, using real tracking data
- **Study type:** simulation feasibility study, not a flight design

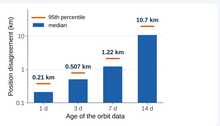
4 How it was tested

- **Computer simulation** of every mission step; each step uses the output of the one before
- **Real data:** 76,045 public tracking records of 962 objects, replayed in time order
- **27 locked experiments**, thousands of random trials each; pass only at **≥ 95%** success

Simulation pipeline

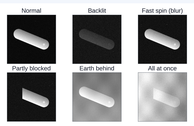
- SGP4 propagation of publication-time GP records; TEME → Earth-fixed frame via GMST
- Monte Carlo errors drawn from measured catalog-disagreement banks
- Rendered images → extended Kalman filter and cylinder-shape estimator
- Protocols frozen with SHA-256 hash + UTC log before outcomes; 72 automated tests

Tracking data ages



Older data, bigger error: 0.2 km after 1 day, 11 km after 2 weeks

Simulated camera



Camera images rendered under 6 lighting and motion conditions

BOTTOM LINE

A reusable space tow truck can work. Fuel is not the problem; travel time, fresh tracking data and a camera that follows the target decide success.

5 · RESULTS: THE MISSION, STEP BY STEP

1 Fly there — *Can it reach the stage?* **8 of 18 within a year**

2 Find it — *Can it spot the stage on arrival?* **98% with fresh data**

3 Close in — *Can it stop safely 30 m away?* **97–98% with a tracking camera**

4 Bring it home — *How much metal comes back?* **7–22 kg of beams per day**

STEP 1 Fly there

Can the servicer reach the stages and bring them back?

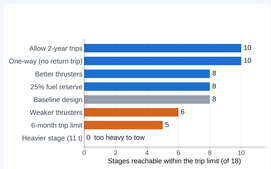
Yes for 8 of 18 stages within a year. Time, not fuel, is the limit.

Round-trip time per stage



Dot = one stage. Below the dashed line = round trip under 1 year.

What changes the count?



One design change per bar. More fuel: no change. Longer trips: +2 stages.

- Each stage circles Earth in its own tilted ring (orbit plane)
- Switching rings directly burns huge fuel, so the servicer **waits** for Earth's bulge to line them up
- The other 10 stages need 1.4–3.8 years of waiting

TECHNICAL DETAIL

- Low-thrust spirals: 0.60 N at I_{sp} = 1,900 s
- 1,350 kg dry, 450 kg xenon (382.5 kg usable)
- 9 t stage added on the return leg
- J_2 plane drift sets the wait; trip ≤ 365 d

$$m_{\text{out}} + m_{\text{ret}} \le (1 - f_r)\, m_{\text{EP}}, \quad T_{\text{thrust}} + T_{\text{wait}} \le 365\ \text{d}$$

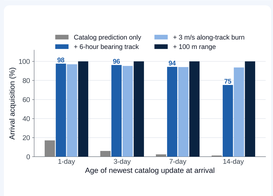
$$\dot{\Omega} = -\tfrac{3}{2} J_2 n \left(\tfrac{R_E}{a}\right)^2 \cos i$$

STEP 2 Find it

When it arrives, can it find the stage?

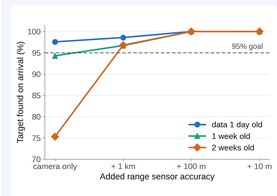
Yes, 98% of the time with 1-day-old tracking data.

Success by data age



Gray: tracking data alone. Dark blue: after 6 hours of camera watching.

Adding a distance sensor



Even a rough (1 km) distance sensor lifts 2-week-old data from 75% to 97%.

- The servicer arrives 1,000 km behind and watches the stage with a camera for 6 hours
- A camera shows **direction, not distance**, so old data hurt: 75% at 2 weeks
- Fix: fresh data, or a simple distance sensor

TECHNICAL DETAIL

- Co-orbital arrival 1,000 km behind the stage
- 5° camera, magnitude 10, 20″ bearing noise
- 37 frames over 6 h → batch least squares
- Prior from the catalog-error bank at that data age
- Hand-off: velocity error ≤ 0.10 m/s

$$\mathbf{b}_k = \frac{\mathbf{r}(t_k)}{\|\mathbf{r}(t_k)\|} + \boldsymbol{\varepsilon}_k, \qquad \mathbf{r}(t_k) = \boldsymbol{\Phi}(t_k)\,\mathbf{x}_0$$

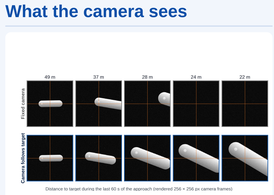
$$\mathbf{P}^{+} = (\mathbf{P}_0^{-1} + \mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1}$$

STEP 3 Close in

Can it get within 30 m without crashing?

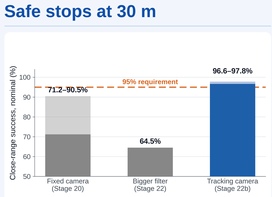
Yes, 97–98% of the time, if the camera turns to follow the stage.

What the camera sees



Top: fixed camera loses the stage at 24 m. Bottom: tracking camera keeps it.

Safe stops at 30 m



Fixed camera: 71–90%, below the 95% goal (dashed). Tracking camera: 97–98%.

- Up close, the 11 m stage fills the picture and slides out of a fixed camera's view; losing sight causes most failures
- Turning the camera toward where the stage should be fixes it

TECHNICAL DETAIL

- 256 px renders: silhouette centroid → bearing, width → range
- 13-frame, 120 s shape estimator; 32° camera inside 120 m
- Hill-equation guidance [3] re-plans at 0, 50, 80, 95%
- Success: within 10 m of the 30 m point at < 0.1 m/s

$$\ddot{x} - 2n\dot{y} - 3n^2 x = a_x, \quad \ddot{y} + 2n\dot{x} = a_y, \quad \ddot{z} + n^2 z = a_z$$

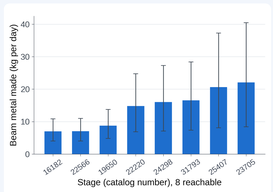
$$\rho \approx \frac{fD}{w_{\text{px}}}, \qquad \text{NEES} = \bar{\mathbf{x}}^T \mathbf{P}^{-1} \bar{\mathbf{x}} \sim \chi^2_6$$

STEP 4 Bring it home

How much metal comes back, and which stage is worth it?

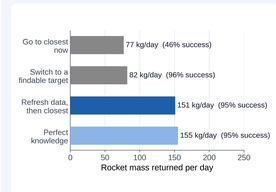
7–22 kg of beam metal per day, about 2–5 t per stage.

Metal per reachable stage



One bar per reachable stage; whiskers show the likely range.

Choosing the next target



With old data, refreshing it first returns 151 kg/day vs 77 kg/day.

- The hub presses the aluminum into beams **without melting it** [5]
- Smart choice matters: waiting for fresh data before choosing **nearly doubles** the metal returned

TECHNICAL DETAIL

- 160,000 Monte Carlo cases: yield, energy, handling time
- Processing 2–12 kWh/kg on a 30 kW hub array
- Best route (NORAD 23705): 22.1 kg/day
- Grasp limit: tumble ≤ 2.3°/s (observed 0.7–4.6°/s [7])

$$R = \frac{m_{\text{target}}\, P_{\text{deliver}}}{T_{\text{elapsed}}}$$

$$\omega_{\max} = \frac{v_{\text{grasp}}}{r} = \frac{0.20\ \text{m/s}}{5\ \text{m}} \approx 2.3^{\circ}/\text{s}$$

6 So, does it work?

In short: reaching, finding and approaching all work; the limits are time and data, not fuel.

- **Yes, in simulation,** for 8 of the 18 stages
- **Fuel is not the problem;** waiting for orbits to line up is
- **Fresh tracking data** decide whether the target is found and which one to pick
- **The camera must follow the target** on the final approach

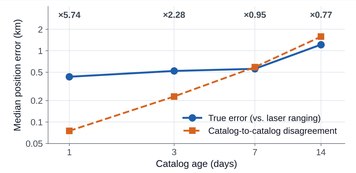
7 How sure are these results?

- Each result comes from **5,000–40,000 random trials**
- A test passes only if even the **pessimistic estimate** (Wilson 95% lower bound, L) is ≥ 95%

$$L = \frac{\hat{p} + \frac{z^2}{2N} - z\sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}}{1 + z^2/N}, \qquad z = 1.96$$

Test	Trials	Success	Pessimistic
Find it (fresh data)	40,000	97.6%	97.4%
Close in (tracking camera)	5,000	97.8%	97.4%
Guidance with good sensors	5,000	99.7%	99.5%

Reality check against lasers



- Tracking data compared with laser measurements of 3 satellites [4]
- Fresh data understate their own error 2–6×
- Re-run with bigger errors: target still found 96% of the time

How good must the sensors be?



- Each square: 5,000 approaches with a given sensor error
- Outlined squares pass: position within **2 m**, speed within **2 cm/s**

8 Limits and next steps

- **What this shows:** the mission architecture works in simulation; flight needs hardware tests of the camera and gripper
- **Grabbing** and stopping a tumbling stage not yet modeled
- **Thin margins:** a 1% chance of navigation-filter failure breaks the 95% goal
- **Next:** test the camera on a spinning model; simulate the grab; measure real tumble rates of all 18 stages

Key references

[1] McKnight et al., Acta Astronautica 181 (2021) 282–291.
[2] Barea et al., Acta Astronautica 170 (2020) 289–301.
[3] Clohessy and Wiltshire, J. Aerospace Sci. 27 (1960) 653–658.
[4] Pearlman et al., J. Geodesy 93 (2019) 2161–2180.
[5] Baker et al., npj Advanced Manufacturing 2 (2025) 29.
[6] Zak, Zenit Stage II, RussianSpaceWeb.
[7] Stromberg et al., Stability analysis of LEO rocket bodies, AMOS 2023.

Data: Space-Track.org (U.S. Space Force); ILRS laser ranging via EDC/DGFI-TUM. Full bibliography (37 sources) in the paper.

[QR: paper and code]